Snort IDS/IPS Lab Implementation

1. Introduction:

Snort is an open-source network security tool used to detect and prevent cyber attacks. It works as both an Intrusion Detection System (IDS) and an Intrusion Prevention System (IPS).An IDS monitors network traffic and generates alerts when suspicious activity is detected, while an IPS not only detects but also blocks malicious traffic in real time.

Snort analyzes network packets using predefined and custom rules. When traffic matches a rule pattern, such as a port scan or malicious request, it generates an alert or takes appropriate action.This project demonstrates how Snort can be configured to monitor network traffic, detect potential attacks using rule-based analysis, and generate alerts for suspicious activity.

IDS detects and alerts on malicious activity, while IPS actively blocks or prevents such traffic.

2. Snort Installation

Install Snort (Ubuntu/Kali)

Open **Terminal** and run some **commands**

2.1 System Update

```
┌──(x369㉿kali)-[~]
└─$ sudo apt update
[sudo] password for x369:
Get:1 http://kali.download/kali kali-rolling InRelease [34.0 kB]
Get:2 http://kali.download/kali kali-rolling/main arm64 Packages [21.0 MB]
Get:3 http://kali.download/kali kali-rolling/main arm64 Contents (deb) [50.1 MB]
Get:4 http://kali.download/kali kali-rolling/contrib arm64 Packages [104 kB]
Get:5 http://kali.download/kali kali-rolling/contrib arm64 Contents (deb) [196 kB]
Get:6 http://kali.download/kali kali-rolling/non-free arm64 Packages [145 kB]
Get:7 http://kali.download/kali kali-rolling/non-free arm64 Contents (deb) [866 kB]
Get:8 http://kali.download/kali kali-rolling/non-free-firmware arm64 Packages [13.4 kB]
Fetched 72.4 MB in 1min 13s (987 kB/s)
1052 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

Fig 2.1 System Update

Updates the system's package list to ensure the latest versions of software are available for installation

2.2 Snort Installation

```
┌──(x369㉿kali)-[~]
└─$ sudo apt install snort -y
snort is already the newest version (3.12.1.0-0kali1).
The following packages were automatically installed and are no longer required:
  amass-common              libnet1                   python3-aiomultiprocess          python3-qasync
  bloodhound.py             libobjc-14-dev            python3-aiosignal                python3-retrying
  debugedit                 libplacebo349             python3-backoff                  python3-serial-asyncio
  dnsmap                    libpocketsphinx3          python3-bluepy                   python3-shodan
  figlet                    libportmidi0              python3-browser-cookie3          python3-shtab
  finger                    libpostproc58             python3-censys                   python3-slowapi
  gir1.2-girepository-2.0   libradare2-5.0.0t64       python3-click-plugins            python3-slugify
  greenbone-feed-sync       librav1e0.7               python3-cmd2                     python3-smmap
  gvmd                      librpmbuild10             python3-frozenlist               python3-spinners
  gvmd-common               librpmsign10              python3-fs                       python3-status
```

Fig 2.2 Snort Installation

Installs the Snort intrusion detection system and its dependencies automatically without prompting for confirmation.

2.3 Version Verification

snort -V

```
┌──(x369㉿kali)-[~]
└─$ snort -v
--------------------------------------------------
o")~   Snort++ 3.12.1.0
--------------------------------------------------
--------------------------------------------------
Network Policy : policy id 1 :
--------------------------------------------------
Inspection Policy : policy id 1 :
--------------------------------------------------
pcap DAQ configured to passive.
--------------------------------------------------
host_cache
    memcap: 33554432 bytes

Snort successfully validated the configuration (with 0 warnings).
o")~   Snort exiting
```

Fig 2.3 Version Verification

Snort was successfully installed and verified using the version command.

2.4 ifconfig

sudo snort -i eth0 -c /etc/snort/snort.lua -A alert_fast

Replace eth0 with your network interface (use **ifconfig**)

```
┌──(x369㉿kali)-[~]
└─$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.2.15  netmask 255.255.255.0  broadcast 10.0.2.255
        inet6 fec0::dc24:69ff:fe67:1cb3  prefixlen 64  scopeid 0x40<site>
        inet6 fe80::dc24:69ff:fe67:1cb3  prefixlen 64  scopeid 0x20<link>
        inet6 fec0::31ae:934c:aa19:4500  prefixlen 64  scopeid 0x40<site>
        ether de:24:69:67:1c:b3  txqueuelen 1000  (Ethernet)
        RX packets 119357  bytes 166430256 (158.7 MiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 21458  bytes 2022777 (1.9 MiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 972  bytes 48680 (47.5 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 972  bytes 48680 (47.5 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

Fig 2.4 ifconfig

Displays and configures network interface details, including IP address, status, and connectivity information.

3. Snort Configuration

3.1 Creating Custom Rule

```
x369@kali: ~
Session  Actions  Edit  View  Help
  GNU nano 8.7                                  /etc/snort/rules/local.rules *
# $Id: local.rules,v 1.11 2004/07/23 20:15:44 bmc Exp $
# ----------------
# LOCAL RULES
# ----------------
# This file intentionally does not come with signatures.  Put your local
# additions here.
alert icmp any any → any any (msg:"Ping Detected"; sid:1000001; rev:1;)
```

Fig 3.1 Custom Rule Configuration

A custom ICMP rule was created to detect ping traffic.

Add this rule :

alert icmp any any -> any any (msg:"Ping Detected"; sid:1000001; rev:1;)

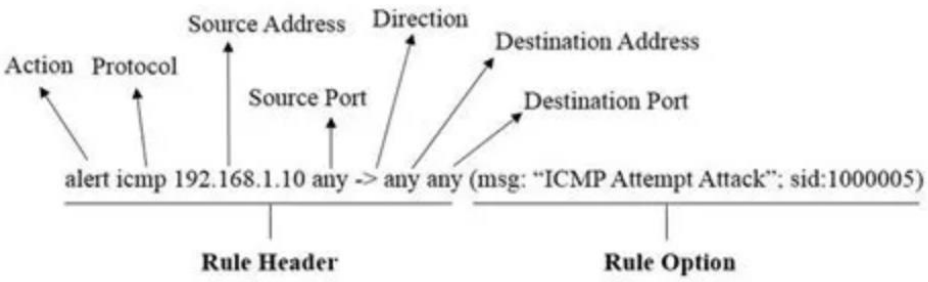

Fig 3.1.1 detailed explanation on command

What this does: Detects **ping (ICMP)** traffic, Generates alert: *Ping Detected*

3.2 Updating Configuration File (snort.lua)

sudo nano /etc/snort/snort.lua

```
ips =
{
    -- use this to enable decoder and inspector alerts
    --enable_builtin_rules = true,

    -- use include for rules files; be sure to set your path
    -- note that rules files can include other rules files
    -- (see also related path vars at the top of snort_defaults.lua)

    variables = default_variables,
    rules = [[
        include /etc/snort/rules/local.rules
    ]]
}
```

Fig 3.2 snort.lua

The Snort configuration file is updated to include the custom local.rules file, enabling detection of user-defined network traffic patterns.

3.3 Running Snort with Rules

sudo snort -i eth0 -c /etc/snort/snort.lua -A alert_fast

```
┌──(x369㉿kali)-[~]
└─$ sudo snort -i eth0 -c /etc/snort/snort.lua -A alert_fast
[sudo] password for x369:
--------------------------------------------------
o")~   Snort++ 3.12.1.0
--------------------------------------------------
Loading /etc/snort/snort.lua:
Loading snort_defaults.lua:
Finished snort_defaults.lua:
        daq
        decode
        stream_ip
        stream_icmp
        stream_tcp
        references
        classifications
        packets
        stream_user
        stream_udp
        stream_file
        arp_spoof
        back_orifice
        dns
        imap
        netflow
```

Fig 3.3 running of snort

Snort is actively monitoring network traffic on the selected interface.

4. Traffic Generation & Detection

Open another terminal and run:

4.1 Generating ICMP Traffic

```
┌──(x369㉿kali)-[~]
└─$ ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=255 time=220 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=255 time=136 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=255 time=83.3 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=255 time=84.7 ms
64 bytes from 8.8.8.8: icmp_seq=5 ttl=255 time=48.8 ms
^C
--- 8.8.8.8 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4014ms
rtt min/avg/max/mdev = 48.848/114.677/220.374/59.762 ms
```

Fig 4.1 ping command

Tests network connectivity by sending ICMP packets to a target host and measuring the response time.

4.2 Alert Generation

```
pcap DAQ configured to passive.
Commencing packet processing
Retry queue interval is: 200 ms
++ [0] eth0
04/12-02:38:40.596019 [**] [1:1000001:1] "Ping Detected" [**] [Priority: 0] {ICMP} 10.0.2.15 -> 8.8.8.8
04/12-02:38:40.816370 [**] [1:1000001:1] "Ping Detected" [**] [Priority: 0] {ICMP} 8.8.8.8 -> 10.0.2.15
04/12-02:38:41.601137 [**] [1:1000001:1] "Ping Detected" [**] [Priority: 0] {ICMP} 10.0.2.15 -> 8.8.8.8
04/12-02:38:41.737358 [**] [1:1000001:1] "Ping Detected" [**] [Priority: 0] {ICMP} 8.8.8.8 -> 10.0.2.15
04/12-02:38:42.602418 [**] [1:1000001:1] "Ping Detected" [**] [Priority: 0] {ICMP} 10.0.2.15 -> 8.8.8.8
04/12-02:38:42.685671 [**] [1:1000001:1] "Ping Detected" [**] [Priority: 0] {ICMP} 8.8.8.8 -> 10.0.2.15
04/12-02:38:43.603692 [**] [1:1000001:1] "Ping Detected" [**] [Priority: 0] {ICMP} 10.0.2.15 -> 8.8.8.8
04/12-02:38:43.688336 [**] [1:1000001:1] "Ping Detected" [**] [Priority: 0] {ICMP} 8.8.8.8 -> 10.0.2.15
04/12-02:38:44.609821 [**] [1:1000001:1] "Ping Detected" [**] [Priority: 0] {ICMP} 10.0.2.15 -> 8.8.8.8
04/12-02:38:44.658657 [**] [1:1000001:1] "Ping Detected" [**] [Priority: 0] {ICMP} 8.8.8.8 -> 10.0.2.15
04/12-02:40:50.087628 [**] [1:1000001:1] "Ping Detected" [**] [Priority: 0] {ICMP} fe80::2 -> ff02::1
04/12-02:40:50.098103 [**] [1:1000001:1] "Ping Detected" [**] [Priority: 0] {ICMP} fe80::dc24:69ff:fe67:1cb3 -> ff02::16
04/12-02:40:50.637463 [**] [1:1000001:1] "Ping Detected" [**] [Priority: 0] {ICMP} fe80::dc24:69ff:fe67:1cb3 -> ff02::16
```

Fig 4.2 logs

Record detected network activities that match defined rules, providing details such as timestamp, source, destination, and type of alert.

5.Traffic Analysis

This experiment demonstrates how Snort monitors network traffic and detects specific patterns using rule-based analysis. The successful alert generation confirms that Snort is effectively identifying ICMP traffic in real time.